

DATE: _09/28/2025_____

GROUP NUMBER: 1

GROUP NAME: _____

GROUP MEMBERS:

Name	Group Role(s)
Dan Caverly	Project Lead
Matthew Drummond	Data Architect
Michael Hicks	Data Scientist
Sean Stack	Communication Liaison

Action Report

- ☒ Action Required
☐ No Action Required

Cause for Action

Complete sentences, concise justification for your team's choice. Include: clarity of goal/purpose, availability of data, team expertise, definition of and likelihood of success, etc.

List any and all methods by which your team came to the decision. This may include personal communication, rubric generation (you may create your own rubric to rank the individual proposals, textbook datasets and/or other project suggestions by teammates), and any other methods you used to come to a group decision

Our team chose to focus on detecting fraudulent credit card transactions because it offered the clearest purpose and had data readily available. The goal is well-defined: build a model that can distinguish between fraudulent and legitimate transactions, which makes the scope manageable and measurable. Given our background, we felt confident in our ability to design and test approaches that could realistically achieve success. To decide, each member listed the pros and cons of each idea, then stated their number one choice of the proposals presented. Fraud detection was chosen as the project we would work on.

Plan-of-Action

List the upcoming, most urgent tasks needed to move forward with the Group Data Project. Make note of the team member roles in these tasks.

Data Preparation: Feature engineering and preprocessing; Split the data into train, calibrate, test split. Apply SMOTE on training set. Matthew Drummond will lead this task.

Model Training: Begin with logistic regression. Move toward xgboost. Explore models of further complexity. Michael Hicks and Dan Caverly will lead this task.

Timelines and Deadlines

Give a timeline and/or Gantt chart.

Data Preparation: Week 6-Week 8

Model Calibration and Training: Week 8 – Week 11

Model Testing: Week 12